Mechatronics Design Laboratory I

ENME 561/661

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Sensors and Transducers

- **Sensor**: an element that produces an output in response to changes in a physical quantity
- **Transducer**: the active element of some sensors that converts the physical quantity into an equivalent electrical signal in the form of current and voltage

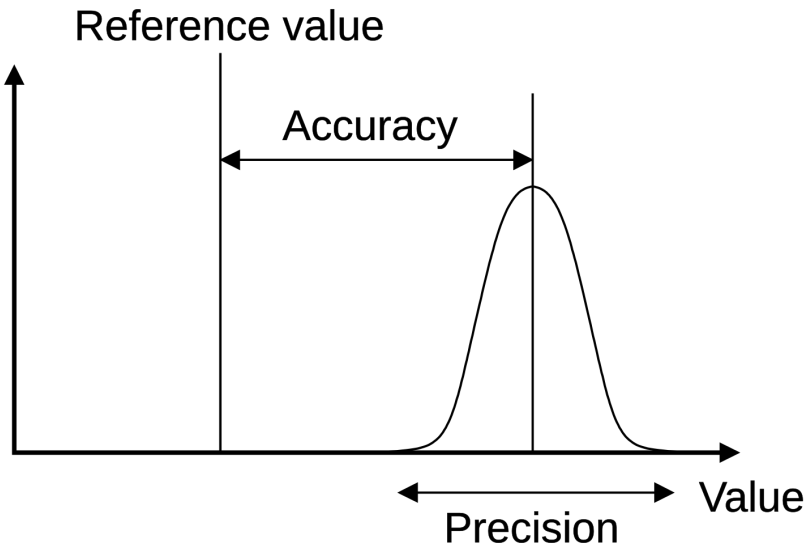
Question: Is a spring scale a transducer?

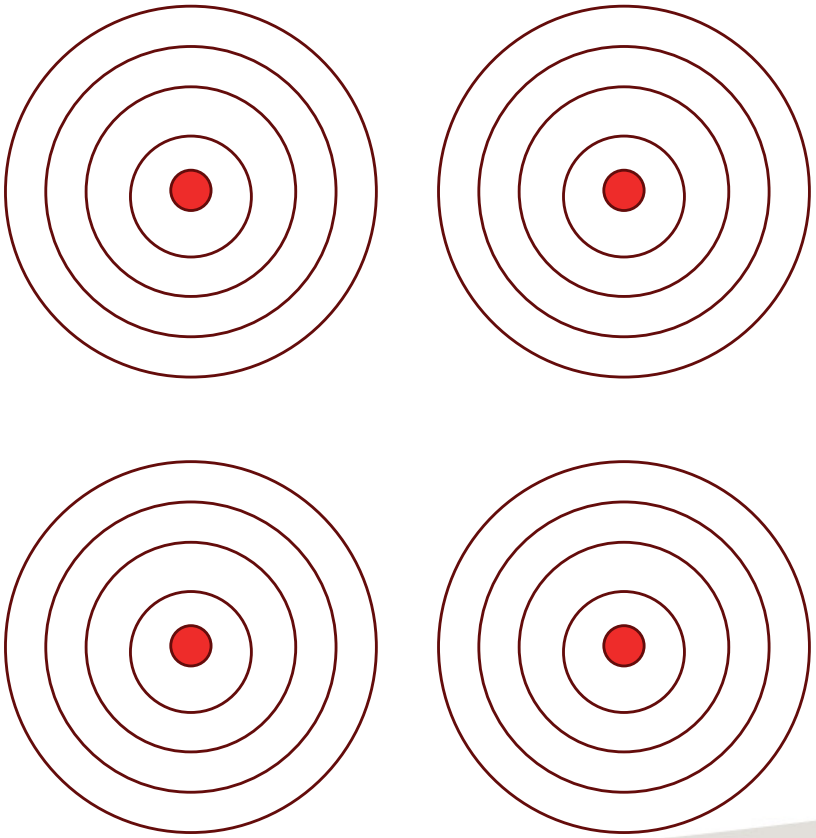


Sensor Performance terminology:

- **Range** is the minimum to the maximum value that can be measured.
- **Resolution** is the smallest change in a measured quantity that an instrument can detect.
  - Resolution is typically expressed in units of the measured
- **Sensitivity** is the relationship between input and output, usually the ratio between the output and the input
- **Hysteresis** is the maximum difference in sensor output for the same input when the input is increasing and when the input is decreasing.

- **Accuracy** refers to the degree to which a measured value corresponds to the true value.
  - Accuracy is typically expressed as a percentage of the full-scale range
- **Precision or Repeatability** is the consistency of measurements, meaning the closeness of agreement between repeated measurements of the same quantity.
  - It's typically expressed as the standard deviation of multiple measurements taken at the same pressure value.

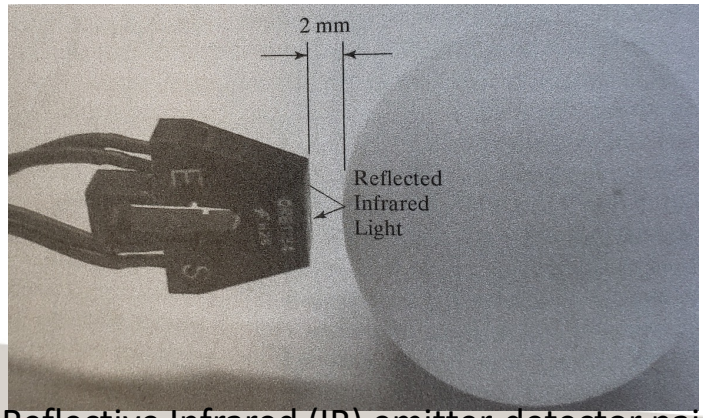




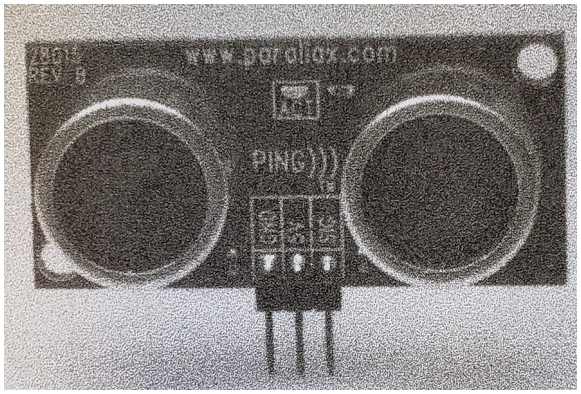
- **Accuracy vs Resolution**

Position and Displacement Sensors

1. Reflective Infrared Sensors
2. Ultrasonic Displacement Sensors
3. Potentiometers
4. Optical Encoders
   - Absolute encoders (to measure angle or position)
   - Incremental encoders (to measure displacement, relative position or angle),



Reflective Infrared (IR) emitter-detector pair



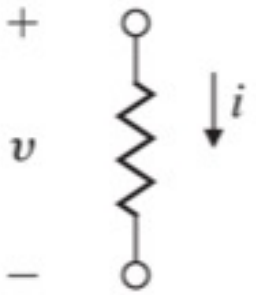
Ultrasonic range finder


Rotary Potentiometer (Pot)

Resistors



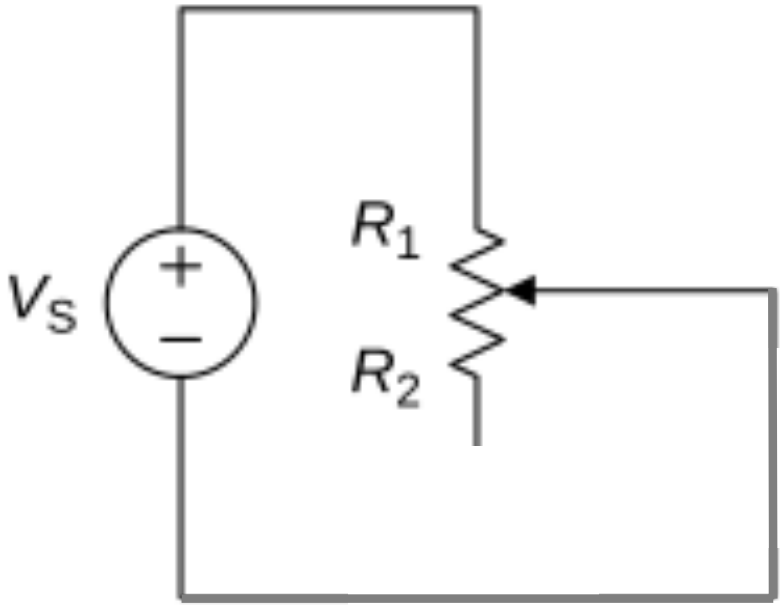
Symbol Equation



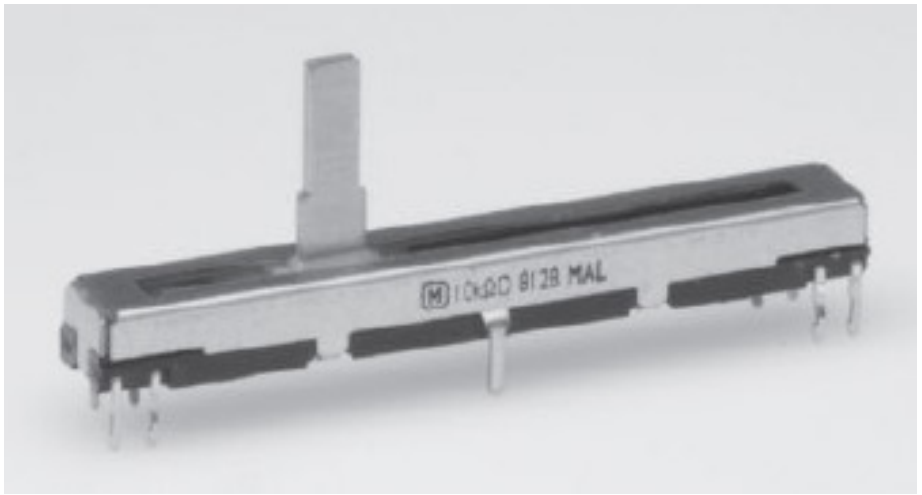
$v = Ri$

Variable resistor

- 2 terminals
- Application: variable low pass frequency filter



Potentiometers



Linear potentiometer



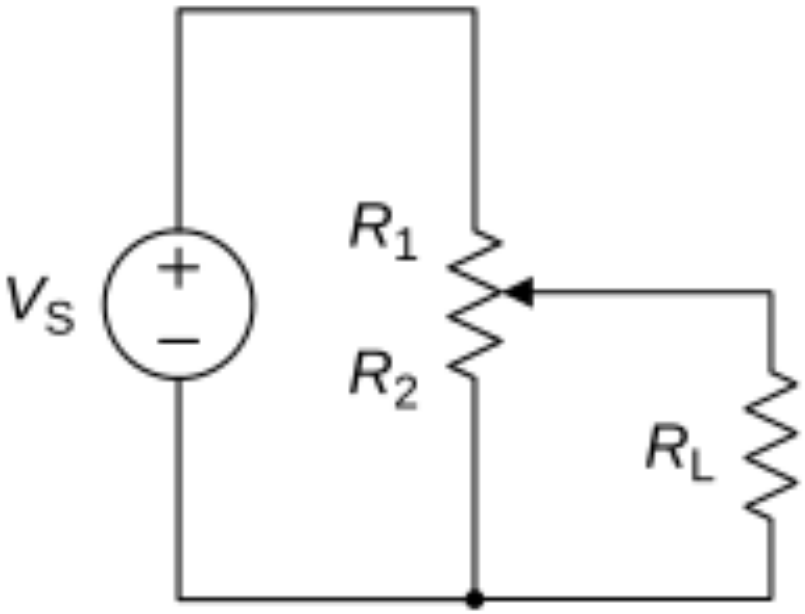
Rotary potentiometer



Symbol

How do potentiometers work?

3 terminals

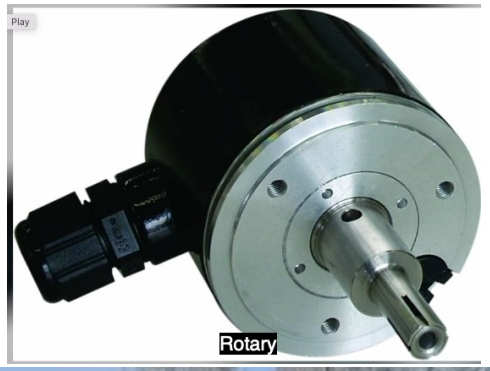




Three pins

Photo Wikipedia

Optical Rotary Encoder



UNIVERSITY OF CALGARY







Robotic arm

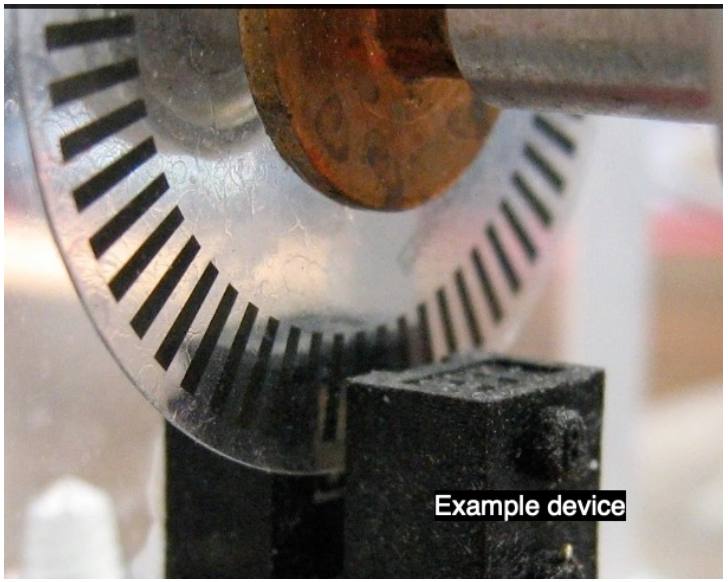




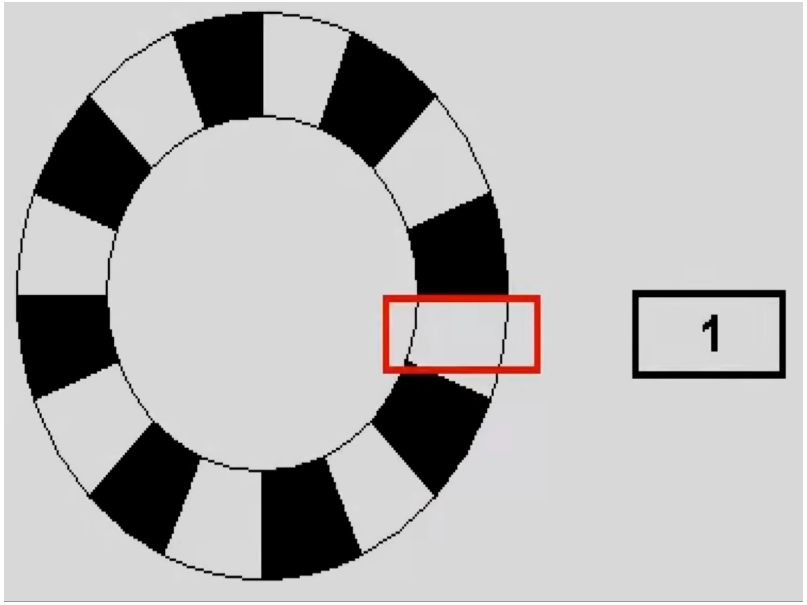
Photos from wikipedia

Optical rotary encoder

- High accuracy while potentiometers affected by friction
- Allows an indefinite number of turns, while potentiometers have a limited range of motion
- Allows measurement of
  - Position/Angle
  - Direction
  - Speed
- 4 elements:
  - Light source (Light Emitting Diode (LED), usually an infrared)
  - Light sensors (Photo-diodes)
  - Optical disk with a pattern of opaque and transparent slots along its periphery (connected to the shaft and rotates with the shaft)
  - Counter

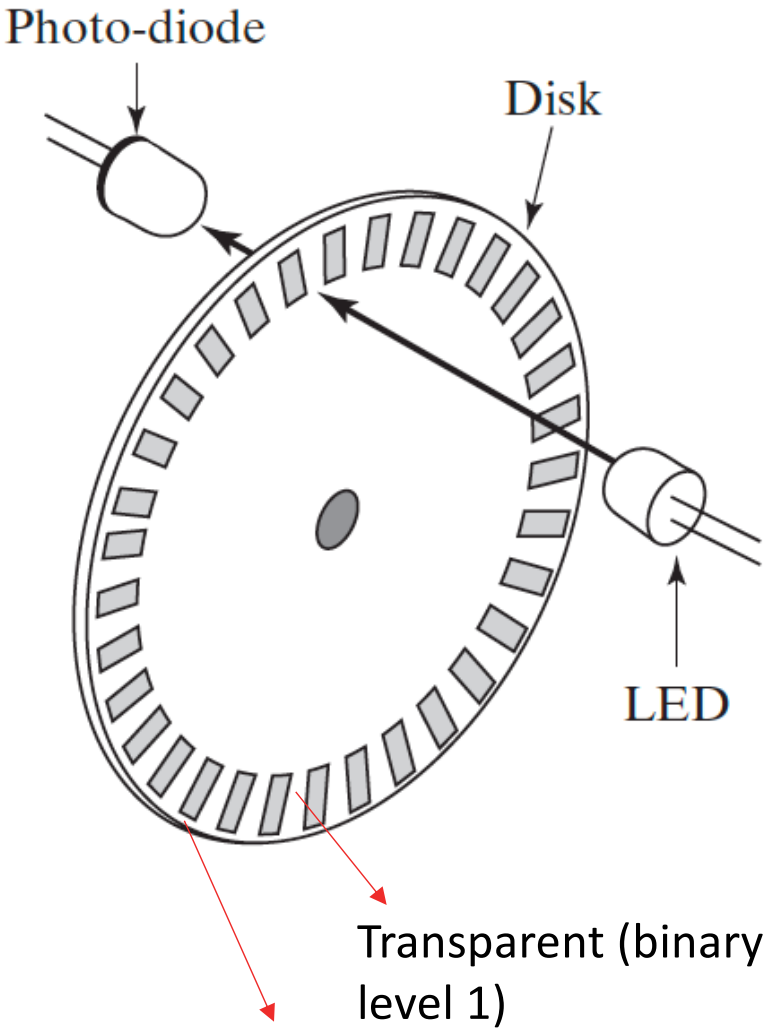


Source: wikipedia



When the disc rotates, the number of pulses generated by the sensor is proportional to the angle that the disc rotates.

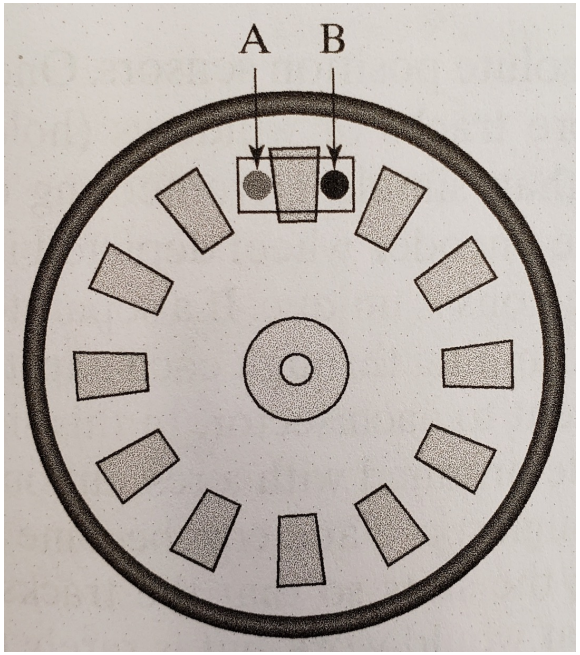
Resolution depends on the number of transparent slots, while the photosensor frequency must be high enough to detect each slot as it passes.



Quadrature encoder

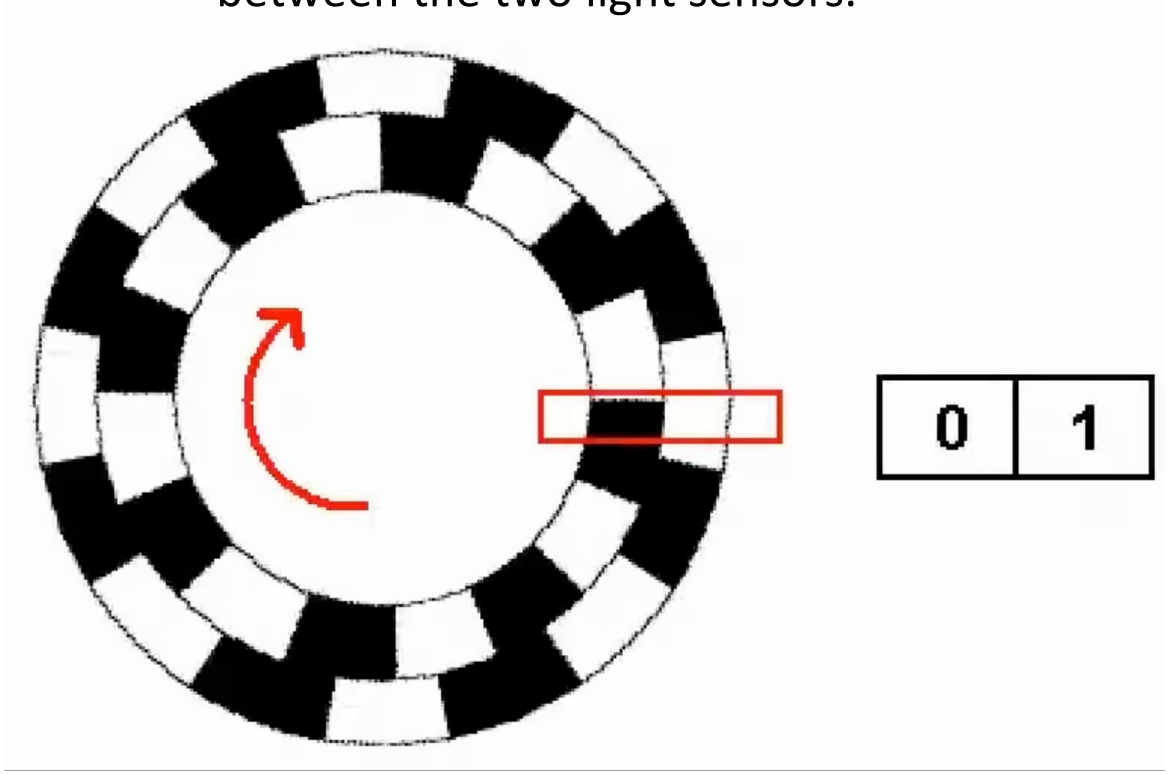
To detect the direction of rotation

One light source on one side and two light sensors on the other side are positioned at half the distance between the slots. There is one track.



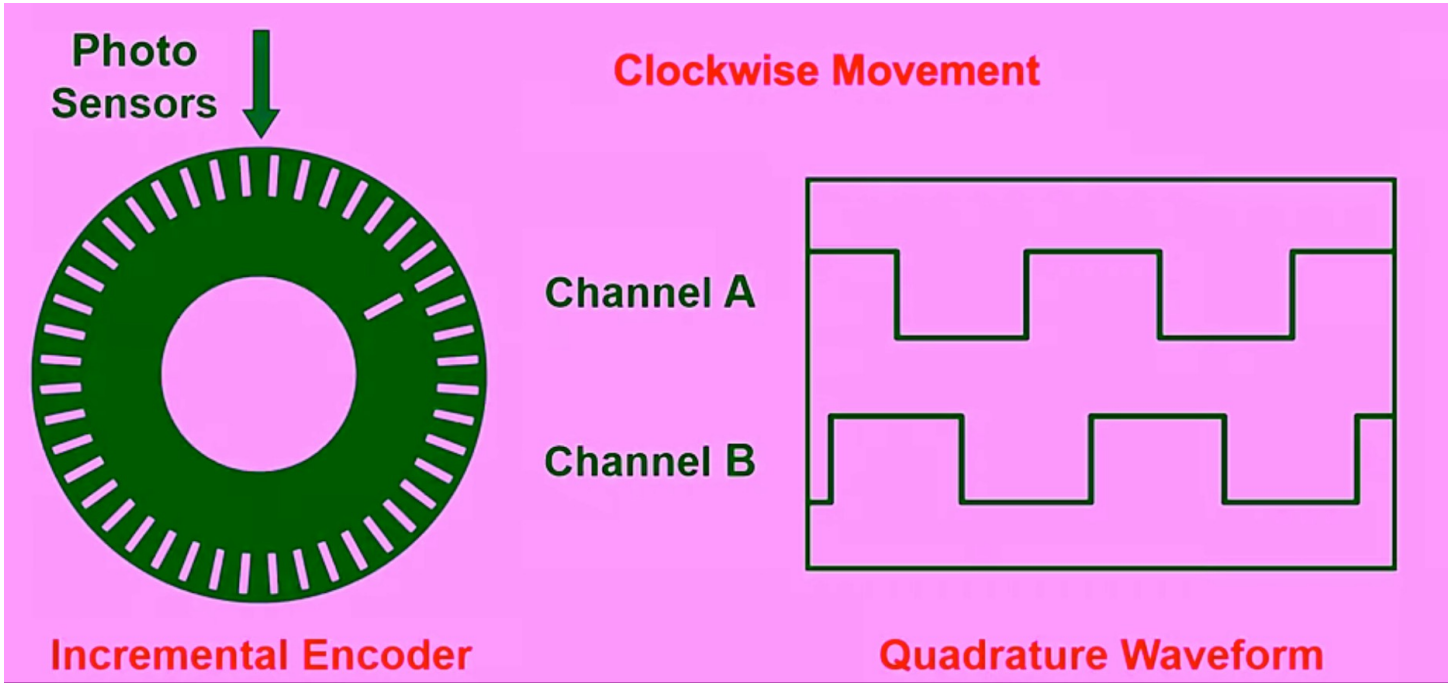
or

Two concentric tracks of lines are arranged with an offset of one-half slot relative to the slots in the other track. There is no offset between the two light sensors.

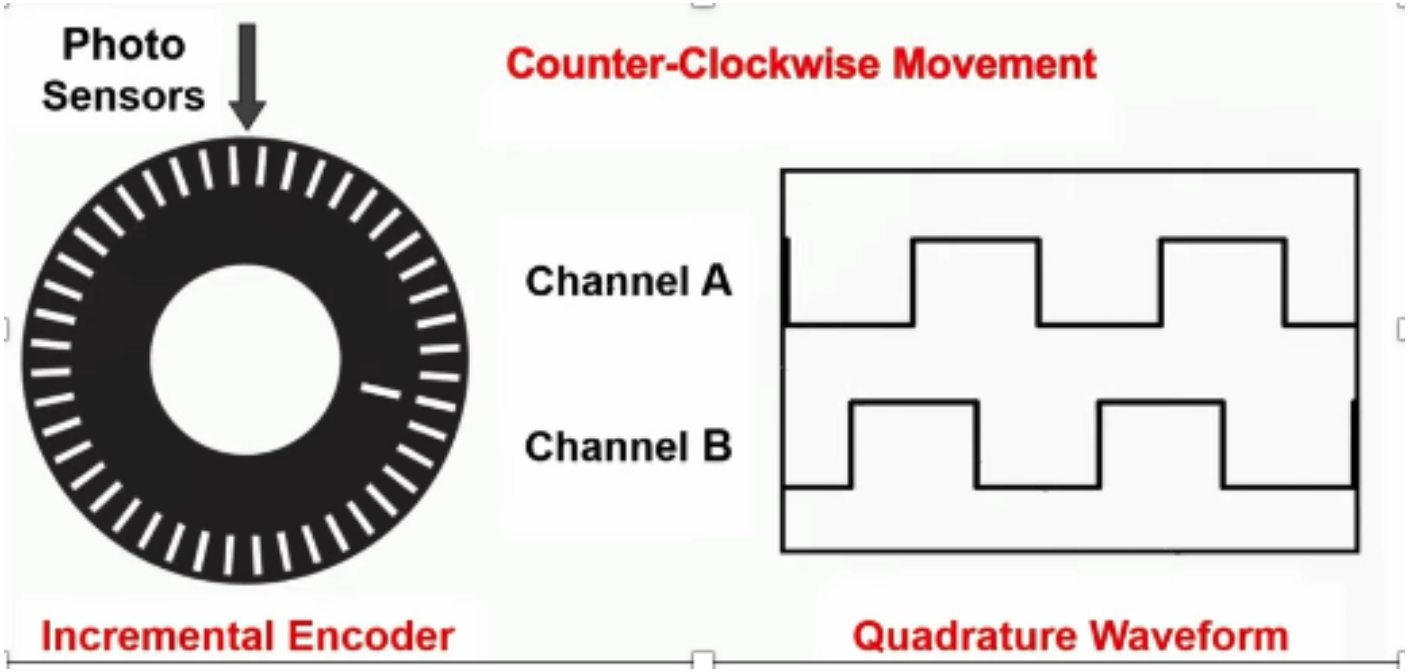


Ref video: Wikipedia

Quadrature Encoder with quadrature index (additional single slot at a fixed location)



Clockwise:
Channel A is leading B

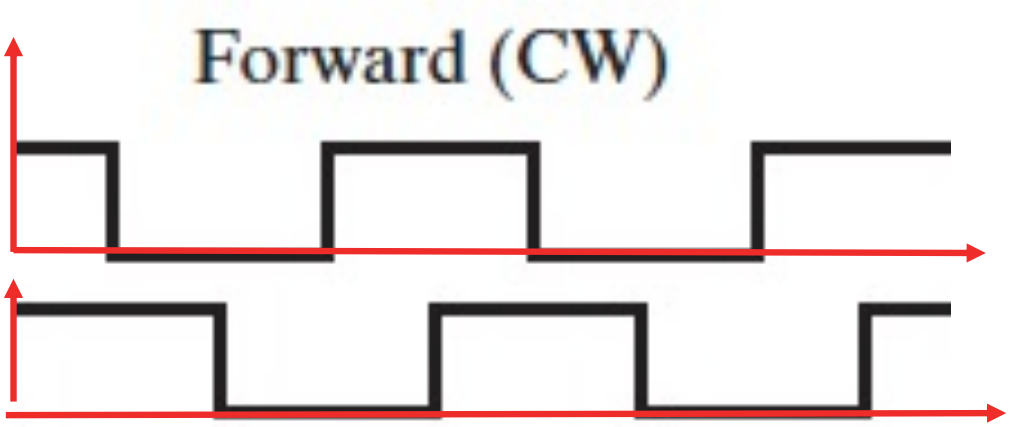


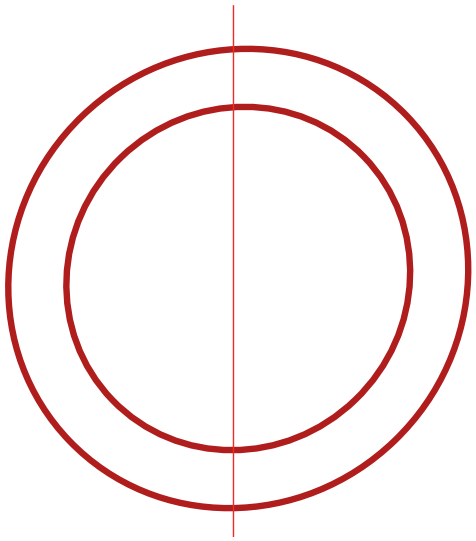
Counter Clockwise:
Channel B is leading A

Ref: Texas Instruments

Quadrature encoder: How it works?

2 tracks



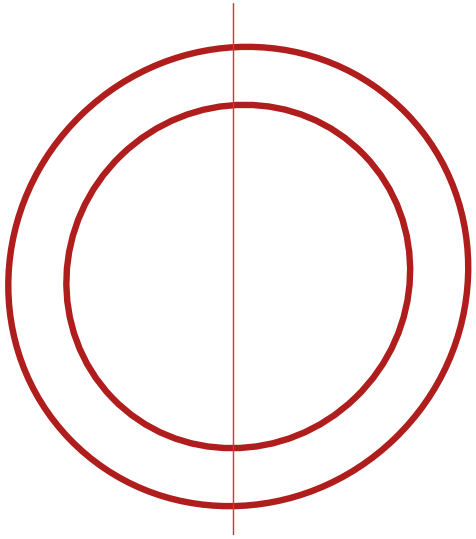


2 concentric circles, offset by ½ width of a slot.

If there are n slots in each track:
Resolution: 360/(n*4)

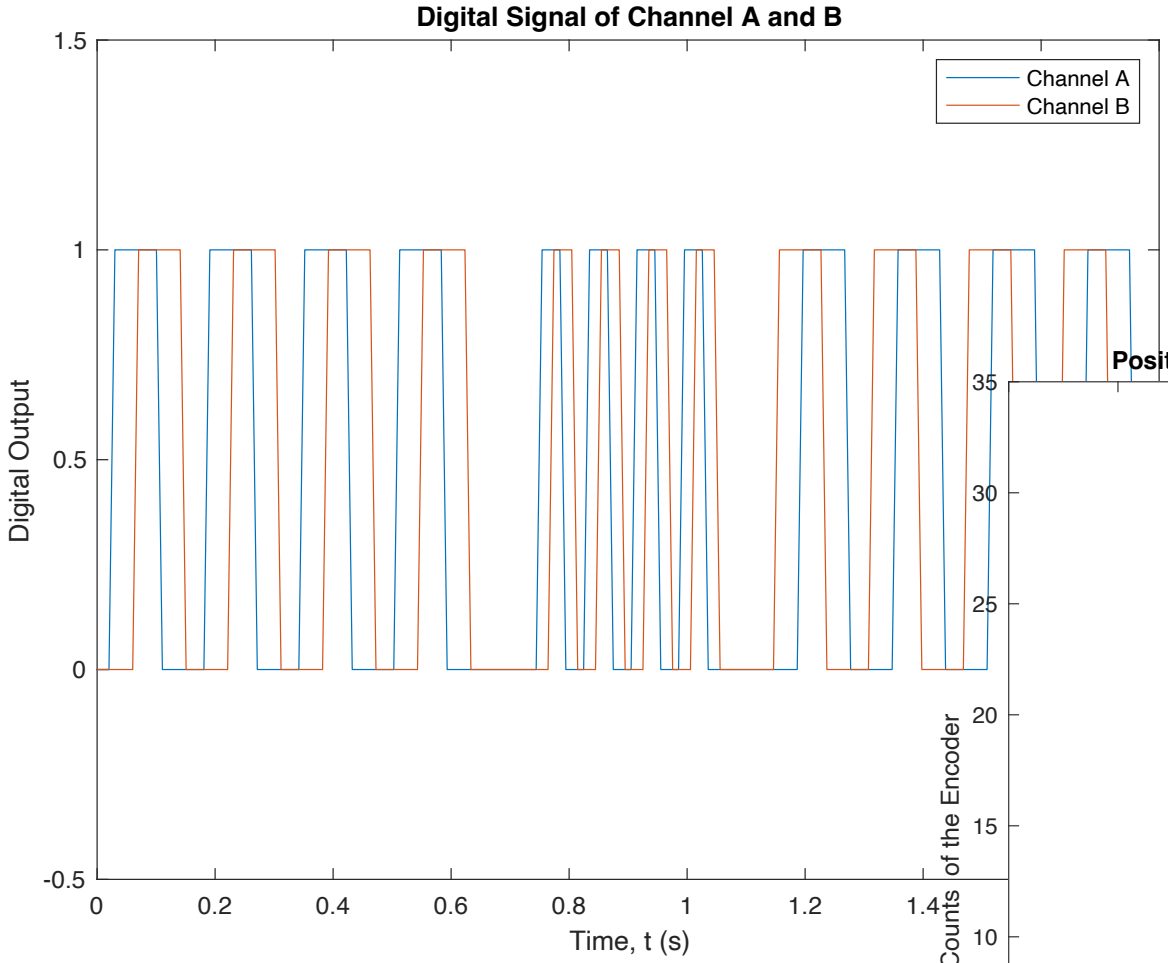
Quadrature encoder: How it works

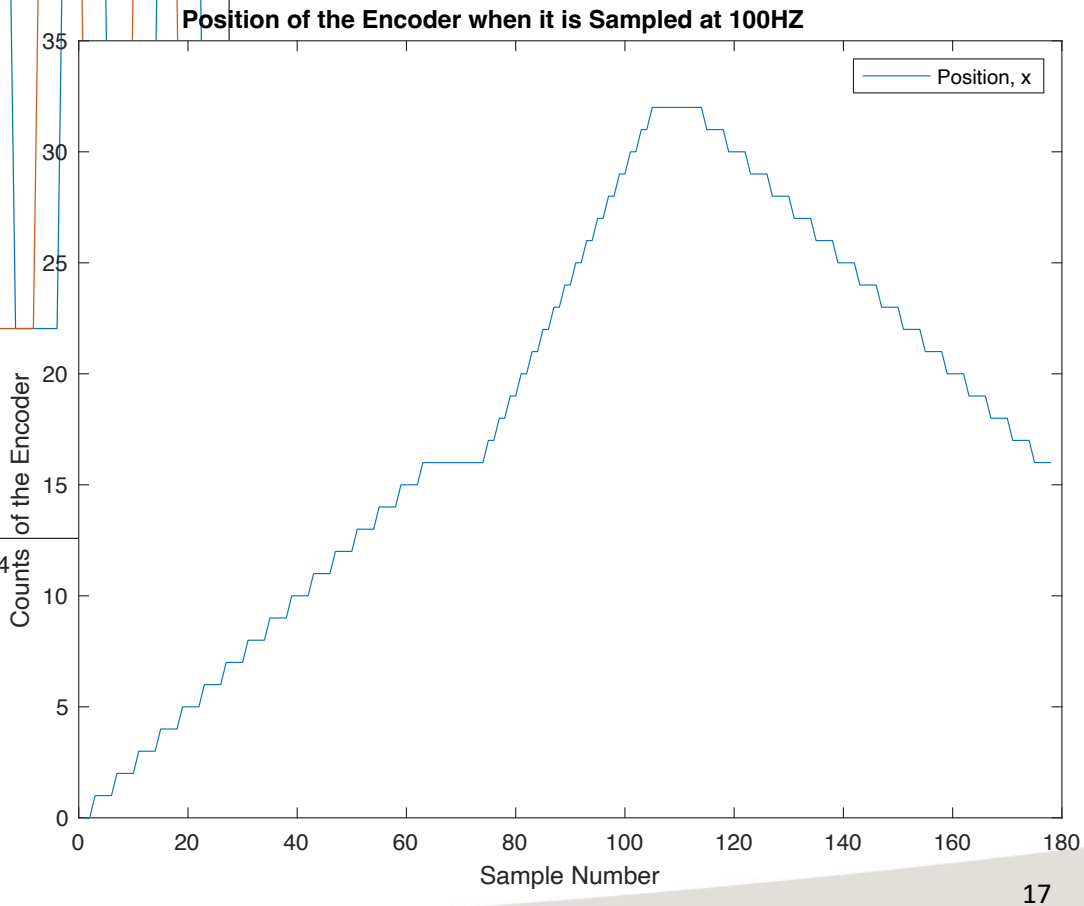
2 tracks



2 concentric circles, offset by ½ width of a slot.

Examples


Digital Signal of Channel A and B
Channel A
Channel B
Digital Output
Time, t (s)
1.5
1
0.5
0
-0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4


Position of the Encoder when it is Sampled at 100HZ
Position, x
Counts of the Encoder
Sample Number
35
30
25
20
15
10
5
0
0
20
40
60
80
100
120
140
160
180

Resolution of Incremental Encoders

Two tracks read in quadrature give four counts per slot

The setup

- n transparent slots in each track
- Two tracks, A and B, offset by a quarter slot pitch
- Electronics count every edge of both channels

Counts per revolution

$$N = 4n$$

Angular resolution

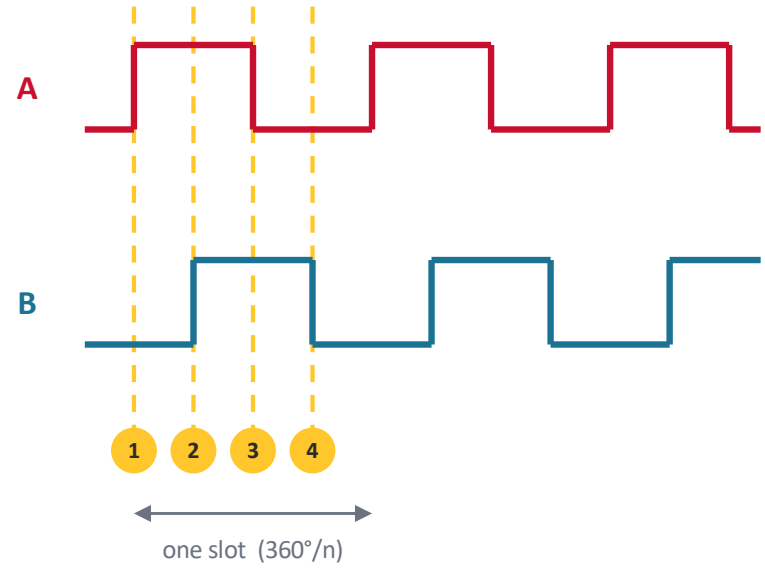
$$\Delta\theta = 360° / 4n$$

Worked example — n = 512

Quantity	Value
Line count in quadrature	4 × 512 = 2048 counts/rev
Angular resolution	360° / 2048 = 0.176°

The inherent resolution is 0.176° — a display showing more digits is not showing more information.

Where the factor of 4 comes from



Four countable edges per slot → N = 4n

More Tracks: Scaling the Resolution

Resolution improves as 2 raised to the number of tracks — until manufacturing stops you

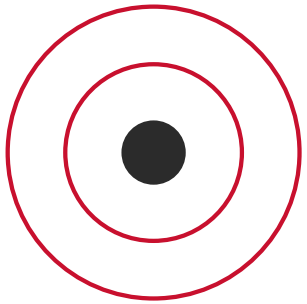
Adding tracks

With T tracks: $N = 2^T \cdot n$ $\Delta\theta = 360° / (2^T \cdot n)$

Tracks T	Counts / rev	Δθ at n = 512
2	$2^2 \cdot n = 4n$	0.176°
4	$2^4 \cdot n = 16n$	0.0439°
8	$2^8 \cdot n = 256n$	0.00275°

The practical limit is manufacturing

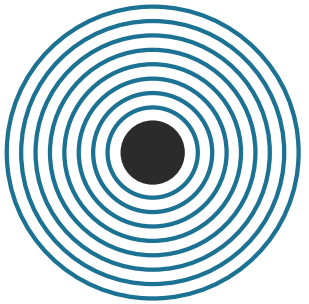
What another track means on the disc



T = 2

2 048 counts/rev

Δθ = 0.176°



T = 8

131 072 counts/rev

Δθ = 0.00275°

Binary code and Grey code

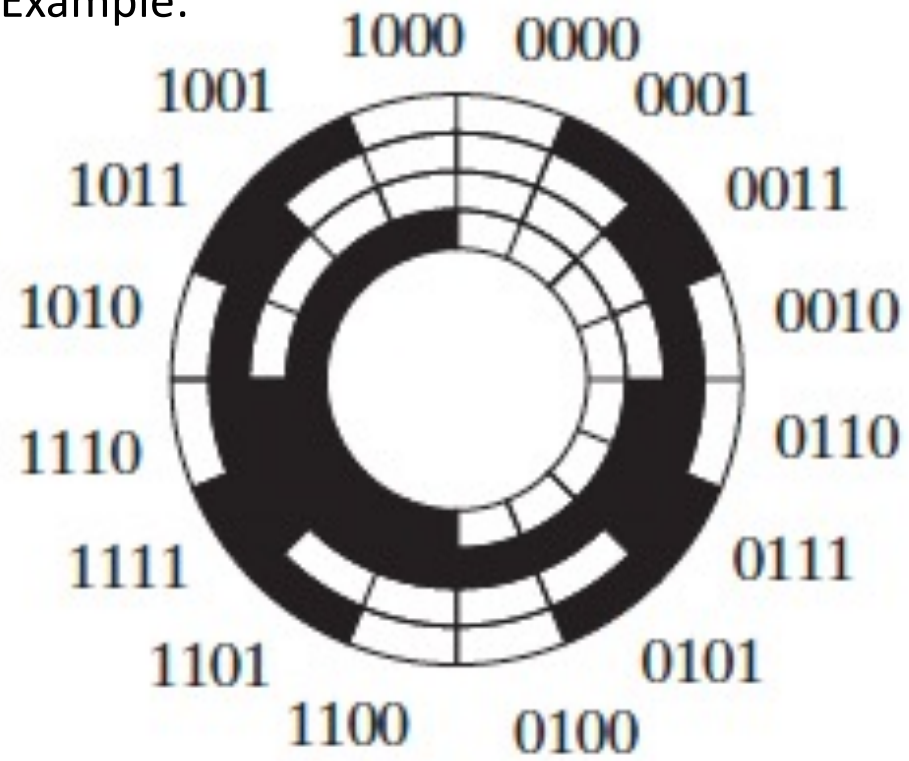
- Both are to represent numbers using only 0s and 1s
- Binary code for performing mathematical operations and storing data in computers. Multiple bits can change between successive numbers.
- Grey code is used when bit-changing transitions are a concern. Only one bit changes between successive values.

Gray code	Binary code
0000	0000
0001	0001
0011	0010
0010	**0011**
0110	**0100**
0111	0101

Absolute optical encoder

Includes variable bar lines on each track, so each spot on the disc has a unique pattern.

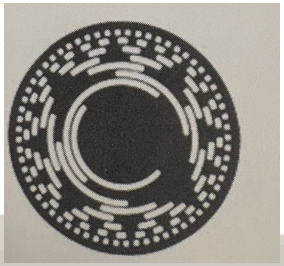
Example:



An encoder wheel with 22.5° resolution (4 bits) i.e. 360/(2^4)

position # (angle)	Gray code	Binary code
0 (0-22.5)	0000	0000
1 (22.5-45)	0001	0001
2 (45-67.5)	0011	0010
3 (67.5-90)	0010	**0011**
4 (90-112.5)	0110	**0100**
5 (112.5-135)	0111	0101
…		

Example of an 8-bit encoder



Reference

- Introduction to mechatronic design, J. Edward Carryer, R. Matthew Ohline, Thomas W. Kenny
- Fundamentals of Mechatronics, 1st Edition, M. Jouaneh
- Texas Instrument
- Image Accuracy and precision illustration, by Pekaje (based on work by Anthony Cutler), via Wikimedia Commons. Licensed under GFDL.